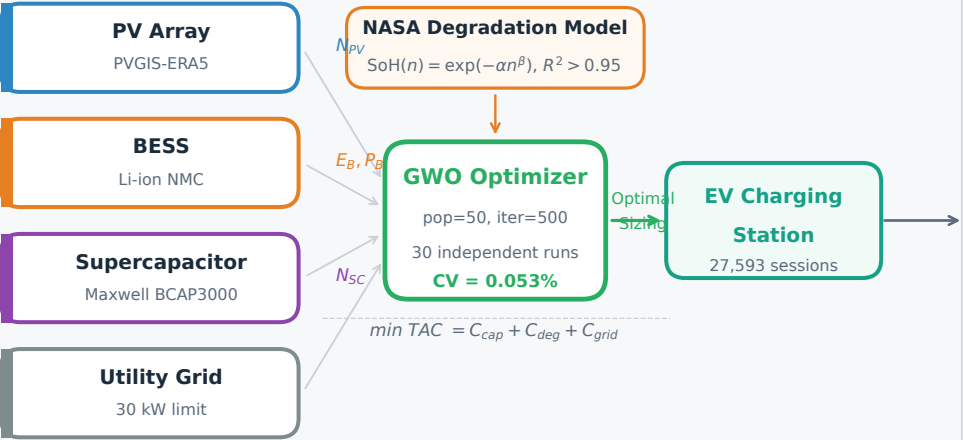


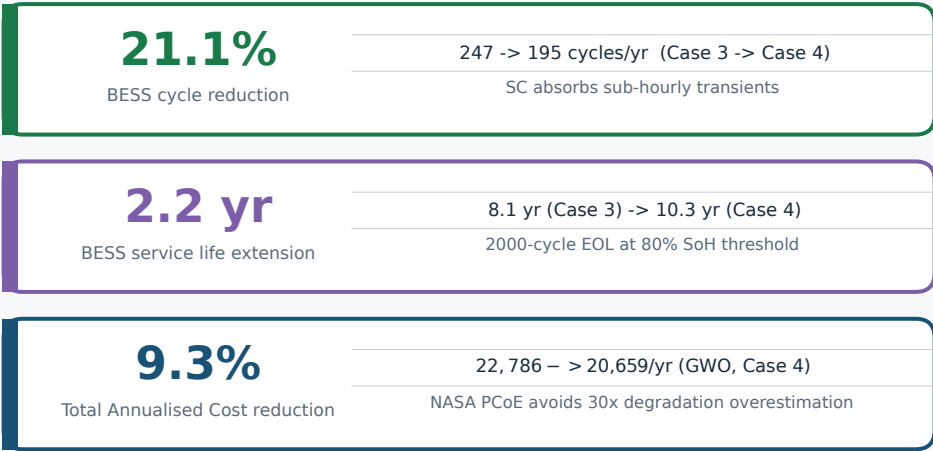
Degradation-Aware Sizing of Hybrid PV-BESS-Supercapacitor Systems for EV Charging Stations

NASA PCoE Calibration | ACN-Data Real Sessions | 30-Run Statistical Validation | Real City Tariffs

DEGRADATION-AWARE SIZING FRAMEWORK



KEY STORAGE RESULTS



SC BANK SIZE IS DRIVEN BY LOCAL DEMAND CHARGE (real tariffs: SCE | Vattenfall | ANEEL)

JPL Pasadena

$N_{SC} = 17 \text{ mod}$

DC: \$25.13/kW/mo

Los Angeles

$N_{SC} = 8 \text{ mod}$

DC: \$25.13/kW/mo

Amsterdam

$N_{SC} = 6 \text{ mod}$

DC: \$8.27/kW/mo

Sao Paulo

$N_{SC} = 6 \text{ mod}$

DC: \$8.05/kW/mo